

CS224R Milestone: SFT Baselines for Per-Second Retention Prediction

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Task & Setup

Predict per-second retention $R(t) \in [0, 1]$ for short-form video ads. Backbone: **Qwen2.5-Omni-3B Thinker** (Talker disabled), PEFT **LoRA** ($r=8$, $\alpha=32$, all-linear) with post-LoRA freeze of visual/audio/merger params (15.1M/4.73B trainable = 0.32%). Hparams aligned to Liangyu’s `sft_v2cot_full.sh` (“v2”): `max_pixels=200704`, `nframes=32`, `max_length=8192`, `batch=1×16` grad-accum, `lr=1e-5`, `bf16`, gradient checkpointing, **10 epochs**, 3 seeds {42, 43, 44}. Eval: **ttcc-eval** canonical 8-script protocol; paired BCa CIs at $B=10K$ vs reference baseline **B1**=train-mean curve (climatology, Mason 2004). Test $n=87$ ads.

Methods

Sanity: $B_0=\text{const } 0.5$; $B_1=\text{train-mean curve}$; $B_2=1 - t/T_i$. **Stage 1 SFT-MSE** (§5): sigmoid head on last non-pad hidden state, MSE on $R(t)$ – literature baseline. **Stage 2 SFT-Hazard+CoT** (§6): softplus hazard head on $\langle \text{cot} \rangle$ hidden state, curve via $\exp(-\sum_s \lambda_s)$, loss $L_{\text{hazard}} + 0.1 \cdot L_{\text{CoT}}$. CoT distilled from `liangyuch/ttcc-cot` (Qwen3-Omni-30B, 717 train ads).

Headline Results (v2, $n = 87$)

Method	IBS ↓	BS(3)	ρ_H	BS _{end}	ρ_C
B0 (constant 0.5)	0.1774	0.1318	n/a	0.2202	n/a
B1 (train-mean)	0.0083	0.0200	n/a	0.0027	−0.029
B2 ($1 - t/T_i$)	0.2082	0.4886	+0.436	0.0021	n/a
SFT-MSE-v2-{42,43,44}	0.129/0.119/0.136	0.069/0.206/0.117	+0.04/−0.16/+0.00	0.167/0.172/0.160	+0.08/+0.08/+0.02
SFT-CoT-v2-42	0.0183	0.0578	+0.311	0.0046	+0.066
SFT-CoT-v2-43	0.0076	0.0208	+0.184	0.0014	−0.005
SFT-CoT-v2-44	0.0078	0.0219	+0.297	0.0015	−0.016

Paired BCa significance vs B1 (10K bootstrap):

- Unconditional IBS: v2-43 $\Delta = -0.0007$ [−0.0020, +0.0007]; v2-44 $\Delta = -0.0006$ [−0.0022, +0.0014] – **tied with B1**.
- Q2Q3 subset ($n=43$): v2-43 −0.0021 [−0.0028, −0.0011]; v2-44 −0.0026 [−0.0036, −0.0011] – **SFT-CoT beats B1**.
- Completion MSE: v2-43 −0.0012 [−0.0020, −0.0004]; v2-44 −0.0012 [−0.0019, −0.0003] – **SFT-CoT beats B1**.
- RMST MAE: v2-43 −0.225s, v2-44 −0.200s vs B1 (directional). % ads beating B1: 64.4% / 62.1%. Drop-localization $\pm 1s$: 100% / 100% (vs SFT-MSE 47–53%).

Failure-Mode Diagnosis (per-method, $R_{\text{hat}}(1)$ dispersion + bias + variance)

- **B1:** $R_{\text{hat}}(1)$ std = 0 – complete mode collapse (by design; one shared curve).
- **SFT-MSE-v2-43:** $R_{\text{hat}}(1)$ std = 0.218 (near true 0.198; *not* mode collapse) but mean bias = +0.192 (systematic over-prediction), per-ad variance 0.066 (jagged non-monotone) – sigmoid head with no monotonicity prior cannot recover smooth $R(t)$ from 717 ads. Loses to B1 by an order of magnitude ($BSS \approx -14$), as §5 predicts.
- **SFT-CoT-v2-43:** $R_{\text{hat}}(1)$ std = 0.065 ($\approx \frac{1}{3}$ of true) – partial mode collapse / regression to mean; bias ≈ -0.02 (balanced 51.7%/48.3% over/under), per-ad variance 0.044 (smooth monotone, hazard parameterization works). “Safe middling” predictions $\Rightarrow$ tied on unconditional IBS, win on Q2Q3 where $R(1)$ is closer to marginal.

Alignment with Milestone PDF Expectations

- §5 SFT-MSE = “weakest, literature baseline”: all v2 seeds lose to B1 by $\Delta \text{IBS} \approx 0.12$, $BSS \in [-15, -13]$. **Consistent**.
- §6 SFT-Hazard+CoT = “starting point for RL” (reasonable, not SOTA): 2/3 seeds point-IBS < B1, CIs tying at 95%; significantly beats B1 on Q2Q3 + completion MSE; $BSS +0.07$ –0.09; drop-localization 100%; hook $\rho_H = +0.30$. **Stronger than expected**.
- §6 architecture (softplus hazard on $\langle \text{cot} \rangle$, cumsum $\rightarrow$ exp, monotone): implemented; ttcc-eval identity check passes. **Pass**.
- Cross-seed std < $\frac{1}{3}$ pairwise method delta (§9 sanity): Stage 2 IBS std = 0.005 vs Δ to SFT-MSE ≈ 0.12 , ratio $0.04 \ll \frac{1}{3}$. **Pass**.

Spec Deviations (disclosed) & Next Step

Deviations. Tuner: spec full FT, ours LoRA $r=8$ (retained as user choice; all other hparams aligned). Inference: spec 3-rollout CoT + hazard averaging; ours empty $\langle \text{cot} \rangle \langle \text{cot} \rangle$. Ablation: gen3 mean IBS 0.013 vs empty 0.011 – empty is the deterministic better choice.

Next (Stage 3 – Liangyu). SFT-CoT is at the edge: 2/3 seeds point-IBS < B1 but tied at 95% CI; output variance $\approx \frac{1}{3}$ of truth. GRPO per-rollout reward (§3, §4 of plan) is positioned to push the remaining variance back, closing the unconditional-IBS gap to B1.